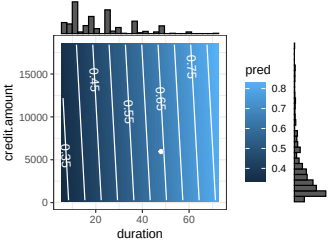
# Interpretable Machine Learning

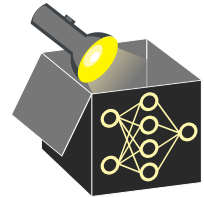
# LIME Examples




**Learning goals**

- See real-world data examples
- See application to image and text data

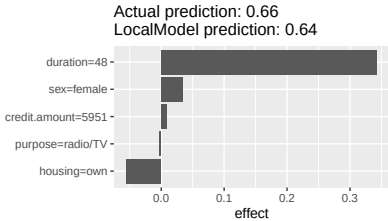
# EXAMPLE ON CREDIT DATASET (TABULAR)



- Model: SVM with RBF kernel
- $\mathbf{x}$: first data point of the dataset with $\hat{f}_{bad}(\mathbf{x}) = 0.658$
- $\mathbf{z}$: training data $\rightsquigarrow$ weighted by the Gower proximity
- Surrogate model $\hat{g}$: $L_1$-regularized linear model with 5 features

| age | sex | job | housing | saving | checking | credit.amount | duration | purpose |
|---|---|---|---|---|---|---|---|---|
| 22 | female | 2 | own | little | moderate | 5951 | 48 | radio/TV |

# EXAMPLE ON CREDIT DATASET (CONT'D)

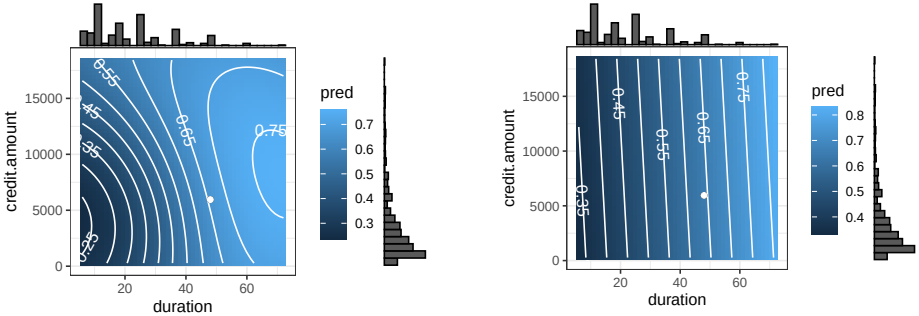



Effects of surrogate model, i.e. $\hat{\theta}^T\mathbf{x}$

- The local model prediction for $\mathbf{x}$ is $\hat{g}(\mathbf{x}) = 0.64$ vs. $\hat{f}(\mathbf{x}) = 0.658$
- $\hat{g}$ has a local fidelity of $L(\hat{f}, \hat{g}, \phi_{\mathbf{x}}) = 4.82$ with $\phi_{\mathbf{x}}(\mathbf{z})$ as the Gower proximity and $L(\hat{f}_{bad}(\mathbf{z}), g(\mathbf{z}))$ as the euclidean distance

# EXAMPLE ON CREDIT DATASET (CONT'D)

- 2-dim ICE plots (aka. prediction surface plot) of credit amount and duration show how the surrogate model $g$ linearly approximates the previously nonlinear prediction surface of $\hat{f}_{bad}$

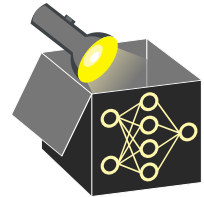



2-dim ICE plot of $\hat{f}_{bad}$ (**left**) and surrogate $g$ (**right**) for features duration and credit amount.

The white dot is **x**. The histograms display the marginal distribution of the training data **X**.

# LIME FOR TEXT DATA

▸ Shen, Ian, (2019)



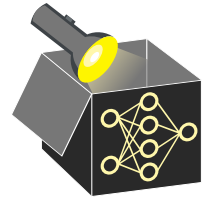
LIME can also be applied to text data:

- Raw text representations:
  - Binary vector indicating the presence or absence of a word
  - A vector of word counts
- Examples for *"This text is the first text."* and *"Finally, this is the last one."*:

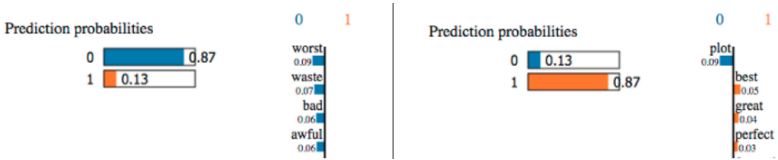
| this | text | is | the | first | finally | last | one |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 1 | 1 | 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | 0 | 1 | 1 | 1 |

- **Sampling**: Randomly set the entry of individual words to 0; equal to removing all occurrences of this word in the text.
- **Proximity**: Exponential kernel with cosine distance.
  - Neglects words that do not occur in both texts
  - Measures the distance irrespective of the text size

# LIME FOR TEXT DATA (CONT'D) ▸ Shen, Ian, (2019)



- Random forest classifier labeling movie reviews from IMDB
  - 0: negative
  - 1: positive
- Surrogate model is a sparse linear model

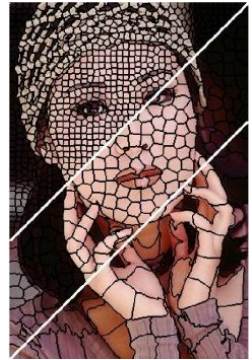



Words like "worst" or "waste" indicate negative review while words like "best" or "great" indicate positive review

# LIME FOR IMAGE DATA

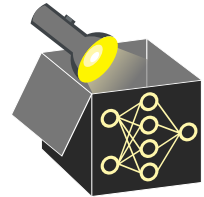
LIME also works for image data:

- **Idea**: Each obs. is represented by a binary vector indicating the presence or absence of superpixels
  ▸ Achanta et al. 2012
- Superpixels are interconnected pixels with similar colors (absence of a single pixel might not have a (strong) effect on the prediction)
- **Warning**: Size of superpixels needs to be determined before the segmentation takes place
- **Sampling**: Randomly switching some of the super pixels “off", i.e., by coloring some superpixels uniformly



Example for superpixels of different sizes

# LIME FOR IMAGE DATA (CONT'D) ▸ Ribeiro. 2016



- Explaining prediction of pre-trained inception neural network classifier
- **Sampling**: Graying out all superpixels besides 10 superpixels
- **Surrogate**: Locally weighted sparse linear models
- **Proximity**: Exponential kernel with euclidean distance

(a) Original Image (b) Explaining *Electric guitar* (c) Explaining *Acoustic guitar* (d) Explaining *Labrador*

Top 3 classes predicted